

ORIGINAL RESEARCH

High Mortality and Complications in Patients Admitted With Takotsubo Cardiomyopathy With More Than Double Mortality in Men Without Improvement in Outcome Over the Years

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BACKGROUND: Takotsubo cardiomyopathy continues to be a major cause of mortality and morbidity. The goal of this study was to evaluate the outcome data of patients with Takotsubo cardiomyopathy using a large inpatient database.

METHODS AND RESULTS: We used the Nationwide Inpatient Sample database for our study. We evaluated trends, mortality, and complications of patients admitted with Takotsubo cardiomyopathy from available years 2016 to 2020 in adults >18. A total of 199 890 patients with Takotsubo were found in our database with 83% being female and higher prevalence with age, White race, and highest income. Mortality was high at 6.5% with no significant improvement over the years studied. Furthermore, major complications were substantial. Cardiogenic shock occurred in 6.6%, atrial fibrillation in 20.7%, cardiac arrest in 3.4%, congestive heart failure in 35.9%, and stroke in 5.3%. Mortality was more than double in men in comparison to women (11.2% versus 5.5%).

CONCLUSIONS: Takotsubo cardiomyopathy is associated with high mortality and complications with no improvement in outcome over the 5-year study with higher mortality in men. Further improvement in care is needed to improve outcomes.

Key Words: apical ballooning syndrome ■ cardiogenic shock ■ cardiomyopathy ■ complications ■ stress induced cardiomyopathy

Takotsubo cardiomyopathy (TC) is a reversible left ventricle dysfunction and is triggered by emotional stress, predominantly in women, or physical stress, predominantly in men.^{1,2} This condition is known to be associated with sex and race disparities and can lead to significant in-hospital mortality and morbidity.^{3,4} Despite the recovery in a later time, little is known about the course and complications of this condition over the recent years. The goal of this study was to use the largest database available to determine the incidence of TC and associated cardiovascular

complications through the years 2016 to 2020 using the Nationwide Inpatient Sample (NIS).

METHODS

Data Source

We used NIS database, which is publicly available for research upon purchase and after completion of educational material about the database. The NIS database includes unweighted hospitalization information for

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CLINICAL PERSPECTIVE

What Is New?

- Using large database, we found that majority of patients with Takotsubo cardiomyopathy (83%) are women with no improvement in mortality of 6.5% over the years as a new finding
- Mortality was more than double in men in comparison to women (11.2% versus 5.5%) with persistent high complications rate of patients with Takotsubo cardiomyopathy such as cardiogenic shock, atrial fibrillation, cardiac arrest, stroke, and congestive heart failure are common complications

What Are the Clinical Implications?

- Clinicians need to improve care of these patients to reduce mortality and study the reason for sex differences in outcome

Nonstandard Abbreviations and Acronyms

NIS	nationwide inpatient sample
TC	Takotsubo cardiomyopathy

about 7 million and weighted data information for about 35 million hospitalizations.⁵ NIS databases are publicly available without any patient's identifier exempt from the institutional review board or need for informed consent.

Study Population and Outcomes

In this retrospective observational cohort study, all patients aged >18 years from the 2016 to 2020 NIS database were included. Our target population was patients hospitalized with TC which was identified using *International Classification of Diseases, Tenth Revision (ICD-10)* code I51.81. The key outcomes of interest including myocardial rupture (I23.2, I23.3, I23.4, I23.5), cardiogenic shock (R57.0), atrial fibrillation (I48), cardiac arrest (I46), congestive heart failure (CHF; I52, I53, I54), and stroke (I60, I61, I62, I63) were also recorded. Patient demographics include age, sex, race or ethnicity, and hospital demographics including median household income, expected primary payer, hospital bed size, location and teaching status of hospital, hospital region, control of hospital along with average hospital length of stay, and average total charges were evaluated.

Statistical Analysis

Patient demographic, clinical, and hospital characteristics are reported as percentage in the [Table](#).

Odds ratios (ORs) and 95% CIs are calculated for continuous variables and proportions and 95% CIs for categorical variables. We studied outcome data over the 5-year period. Categorical outcomes were assessed using chi-square analysis. We performed multivariate logistic regression analysis adjusting for baseline characteristics and known common cardiovascular and pulmonary risk factors for mortality. Multivariable linear regression models were constructed to assess the relationship between the independent variables and length of stay by reporting calculated beta coefficients with 95% CI for each factor. All analyses were conducted following the implementation of population discharge weights. For data sets using NIS, weights are calculated to ensure the sample accurately represents the entire US inpatient population. Hospital identifier as the primary sampling unit and applies discharge weight as the weight variable, which scales each discharge to represent the national population. By applying weights, regression coefficients reflect the relationships at the population level rather than just within the sampled cases. Additionally, variance estimates are adjusted for the survey design, which affects SEs and CIs. In survey-weighted multivariate regressions, the SEs are robust. The survey design settings account for complex patterns in the data, such as clustering within hospitals and stratification, without assuming homoscedasticity (constant variance of errors). All *P* values were 2-sided and *P*<0.05 was considered statistically significant. Data were analyzed using STATA 17 (Stata Corporation, College Station, TX).

RESULTS

Baseline Characteristics

In the investigation of the NIS database from 2016 to 2020, we studied a weighted total population of 148 767 787 (unweighted 29 753 566) with a weighted total of 199 890 (unweighted 39 984) patients who were hospitalized with TC were identified based on the *ICD-10* codes. The annual incidence of patients with TC did not show a consistent pattern but overall increased from 39 015 (unweighted 7803) in 2016 to 41 290 (unweighted 8258) in 2020. The annual incidence of TC based on sex over the years can be seen in [Table](#).

The baseline characteristics are presented in [Table](#). The mean age was 67.09±14.15 years, and 83% of cases were female. As shown in [Figure 1](#), incidence increased from 0.19% to 0.21% for women and from 0.05% to 0.07% of all hospitalizations for men, respectively, during the study years.

The age group >61 consistently had the highest incidence rates. The TC incidence increased across all age groups from 2016 to 2020 shown in [Figure 2](#).

Table. Baseline Characteristics and In-Hospital Complications

	Takotsubo	Others	Total	P-value	Odds ratio (CI)
No. (%)	199 920	148 567 866	148 767 786	...	...
Age,* y	67.09±14.15	57.99±20.24	58.00±20.23	<0.0001	...
Male sex, %	17%	42.72%	42.69%	...	REF
Female sex, %	83%	57.28%	57.31%	<0.0001	3.54 (3.44–3.65)
Race or ethnicity, %					
White	80.39%	66.98%	67.00%	<0.0001	REF
Black	8.20%	15.32%	15.31%	<0.0001	0.45 (0.43–0.46)
Hispanic	6.38%	11.27%	11.27%	<0.0001	0.47 (0.45–0.49)
Asian/Pacific Islander	2.12%	2.77%	2.77%	<0.0001	0.64 (0.59–0.69)
Native American	0.64%	0.66%	0.66%	<0.0001	0.81 (0.70–0.93)
Other†	2.27%	3.00%	2.99%	<0.0001	0.63 (0.59–0.68)
In-hospital complications					
Mortality	6.58%	2.40%	2.41%		2.86 (2.73–2.99)
Myocardial rupture	0.02%	0.003%	0.003%	<0.0001	5.79 (2.74–12.20)
Cardiogenic shock	6.66%	0.56%	0.57%	<0.0001	12.71 (12.19–13.26)
Atrial fibrillation	20.79%	15.48%	15.49%	<0.0001	1.43 (1.40–1.47)
Cardiac arrest	3.42%	0.74%	0.74%	<0.0001	4.79 (4.53–5.06)
Congestive heart failure	35.93%	13.74%	13.77%	<0.0001	3.52 (3.44–3.60)
Stroke	5.38%	2.76%	2.77%	<0.0001	2.00 (1.90–2.10)

*Age, y: mean±SD.

†Includes multiracial.

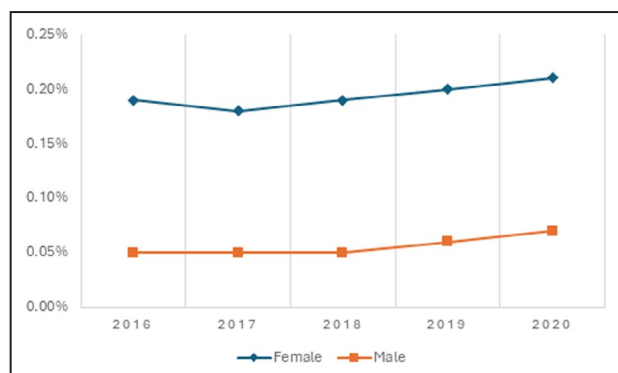
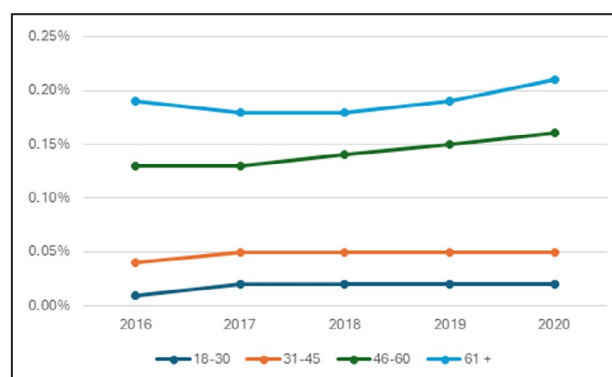
White people had the highest rate of TC accounting for 0.16% of the total population studied followed by Native Americans at 0.13%, whereas the lowest rate was among the Black population of 0.07%. The data revealed noticeable racial disparities in incidence rates from 2016 to 2020 (Figure 3).

Socioeconomic factors including median household income, hospital bed size, and primary payer varied significantly. Patients with TC tend to have higher household incomes and larger hospital bed sizes. Most patients were on Medicare (0.18% versus 0.08% for Medicaid or no charge). Among ownership hospitals, TC rates were highest in private, nonprofit (0.14%), and among teaching hospitals, urban teaching hospitals had the highest rate (0.14%).

In-Hospital Complications

In order to prevent any bias in selecting specific group, we decided to use the entire hospitalized population for comparison for outcomes and mortality. Patients with TC had a high burden of in-hospital mortality (6.58%) and cardiovascular complications including myocardial rupture (0.02%), cardiogenic shock (6.66%), atrial fibrillation (20.79%), cardiac arrest (3.42%), CHF (35.93%), and stroke (5.38%) compared with patients without TC. The OR for in-hospital complications is shown in Table.

Overall mortality resulting from TC was significantly high at 6.58%, compared with 2.41% in patients without TC, with an OR of 2.86 (95% CI, 2.73–2.99; $P<0.001$).

**Figure 1. Takotsubo cardiomyopathy incidence by sex.****Figure 2. Takotsubo cardiomyopathy overall incidence (2016–2020).**

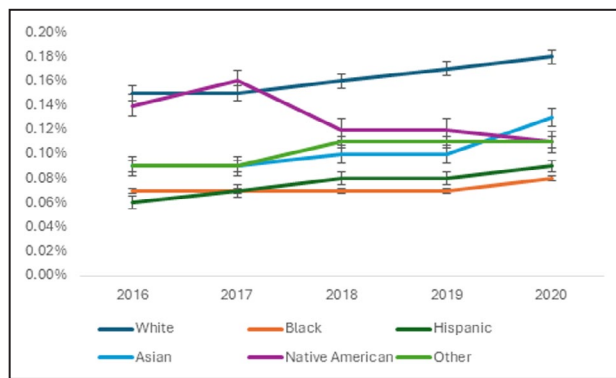


Figure 3. Takotsubo cardiomyopathy incidence over years based on race or ethnicity.

Other includes multiracial.

Patients with TC showed significantly higher odds of cardiogenic shock (OR, 12.71 [95% CI, 12.19–13.26]; $P<0.001$), cardiac arrest (OR, 4.79 [95% CI, 4.53–5.06]; $P<0.001$), CHF (OR, 3.52 [95% CI, 3.44–3.60] $P<0.001$), and myocardial rupture (OR, 5.79 [95% CI, 2.74–12.20]; $P<0.001$) as presented in Figure 4. In addition, patients with TC were 2 times more likely to experience a stroke (OR, 2.00 [95% CI, 1.90–2.10]; $P<0.001$) and had a slightly higher incidence of atrial fibrillation (OR, 1.43 [95% CI, 1.40–1.47]; $P<0.001$).

Over the years of the study, certain complications such as mortality, cardiogenic shock, cardiac arrest, CHF, and stroke have shown an apparent increase, whereas myocardial rupture and atrial fibrillation incidence remained relatively stable. Mortality increased from 5.63% to 8.38%, cardiogenic shock from 5.81% to 7.41%, cardiac arrest from 2.64% to 4.47%, CHF from 34.77% to 37.61%, and stroke from 4.97% to 5.9%, from 2016 to 2020, respectively. Atrial fibrillation rate steadily escalated from 21.04% in 2016 to 21.31% in 2020. Myocardial rupture rate was very low at 0.02%. The most common complication was CHF followed by atrial fibrillation. Figure 5 showing mortality over time based on sex with stable mortality in male and female sexes over the years but almost double the mortality in men in comparison to female sex. Using multivariate analysis adjusting for age, sex, race or ethnicity, chronic obstructive lung disease, smoking, chronic kidney disease,

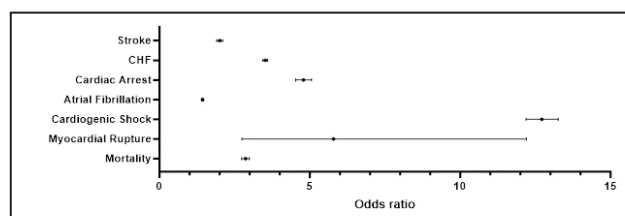


Figure 4. Forest plot showing complications related to Takotsubo cardiomyopathy.

CHF indicates congestive heart failure.

hypertension, hyperlipidemia and diabetes, TC remained independently associated with all-cause mortality (OR, 2.67 [95% CI, 2.5–2.8], $P<0.001$).

Length of Stay and Average Total Charges

The median length of stay for patients admitted with TC remained consistent across the study period, with a median of 4 days from 2016 to 2020. Based on multivariable linear regression, factors significantly associated with longer length of stay compared with reference included Black race ($\beta=1.53$, $P<0.011$), Asian race ($\beta=2.85$, $P<0.001$), Native American race ($\beta=1.54$, $P=0.03$), medium ($\beta=0.34$, $P<0.001$) and large hospital bed size ($\beta=1.42$, $P<0.001$), and Medicaid insurance ($\beta=0.86$, $P<0.001$). Conversely, older age (31–45 years: $\beta=-2.52$, $P<0.001$; 46–60 years: $\beta=-3.11$, $P<0.001$; 61+ years: $\beta=-3.56$, $P<0.001$), female sex ($\beta=-2.31$, $P<0.001$), treatment in the Midwest ($\beta=-1.03$, $P<0.001$), South ($\beta=-0.9$, $P<0.001$), or West ($\beta=-1.4$, $P<0.001$) regions, private insurance ($\beta=-0.54$, $P<0.001$), and private, not-for-profit hospital ownership ($\beta=-1.63$, $P<0.001$) were associated with shorter length of stay.

DISCUSSION

Our study of 199920 patients hospitalized with TC in a large inpatient database from 2016 to 2020 revealed major insights into the epidemiology and complications of this condition over the years. Figure 6 summarizes our study.

Our study discovered a new age-related pattern in patients with TC, identifying a critical turning point. It has been previously found that TC incidence increases with aging and usually follows a stressful situation, either emotional or physical.⁶ Our study revealed a sudden increase in incidence in the 46 to 60 years age group compared with the 31 to 45 years age group, with the older age group showing 2.6 to 3.25 times higher incidence. This sudden rise could be due to a combination of increased stress levels, hormonal variations, and the onset of cardiovascular risk factors along with under-treatment for hypertension and hyperlipidemia, alcohol use, and smoking.^{7–9} Although our data did not differentiate between 4 various types of TC¹⁰ due to the use of a single ICD-10 code, the recently diagnosed type of TC known as reverse-type TC tends to occur mainly in younger individuals.¹⁰ The higher percentage of men in the current study might also contribute to this new age cutoff, as men tend to present with TC at a younger age than women.^{3,11} This new age-related finding holds important clinical implications. It could serve as a useful tool in discriminating between acute coronary syndrome and TC, aiding in the early diagnosis of this condition specific to older adults.

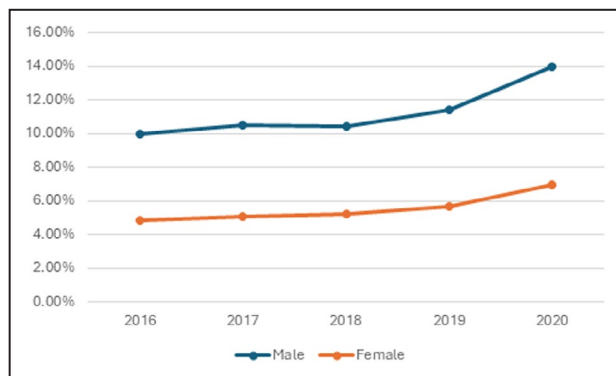


Figure 5. Takotsubo cardiomyopathy mortality by sex.

Consistent with previous studies,⁸ female sex was more likely to present with TC. The reason for this predominance is not fully understood. Hormonal changes especially in older ages might play a role.^{12,13} In a study by Ueyama et al., estradiol decreased the pathological changes in heart rate caused by emotional stress in rats.¹⁴ However, the percentage of women in our study was lower (83%) compared with as high of 90%,^{10,12} 97%,¹⁵ and 89%¹⁶ in previous studies. This change might reflect previous underdiagnosis TC in men¹ and increased awareness of the condition in men.

Our observation of race showed that the lowest hospitalization for TC was among White people in 2016 and the highest in 2020 among White people followed by Native American people. compatible with our results, in the study by Brinjkji et al., the majority of patients with TC were White.¹⁶ In contrast to our results,

Native American people had the lowest frequency of TC using NIS data in 2008.⁸ Consistent with our results, White hospitalization increased over 2006 to 2014 in a nationwide sample study.⁴ This finding might be related to better recognition of TC across all races, especially White people.

Our results are compatible with prior studies in the United States regarding increased TC-associated hospitalization.^{4,17} It remains unclear whether the rise happened in the actual incidence, the number of diagnoses due to greater clinician awareness of the condition,¹⁸ or more frequent use of different diagnostic criteria, such as Mayo Clinic criteria.¹⁹ Despite the studies on TC since its discovery in 1991 by Dote et al.,²⁰ this condition is still not well understood from many perspectives. Although catecholamine-induced myocardial injury is the most accepted mechanism for TC, the exact pathophysiology remains unknown.¹² Therefore, due to a completely poor understanding of pathophysiology, despite rigorous advancements in cardiac care, the treatment options in the acute TC phase are not completely determined and mortality of TC is still high. Mortality rates have been estimated at 0% to 8% in different studies,¹ showing an incremental pattern over the years. Brinjkji et al. reported an in-hospital mortality rate of 4.2% for patients with TC from 2008 to 2009.¹⁶ In a review by Singh et al. included 37 research papers from different regions, and overall in-hospital mortality was estimated at 4.5%.²¹ In the Mayo Clinic Takotsubo syndrome registry, in-hospital mortality was 3%.²² An interesting finding revealed by our study was that not only did mortality rates increase over the years of study but also it was as high as 8.3%

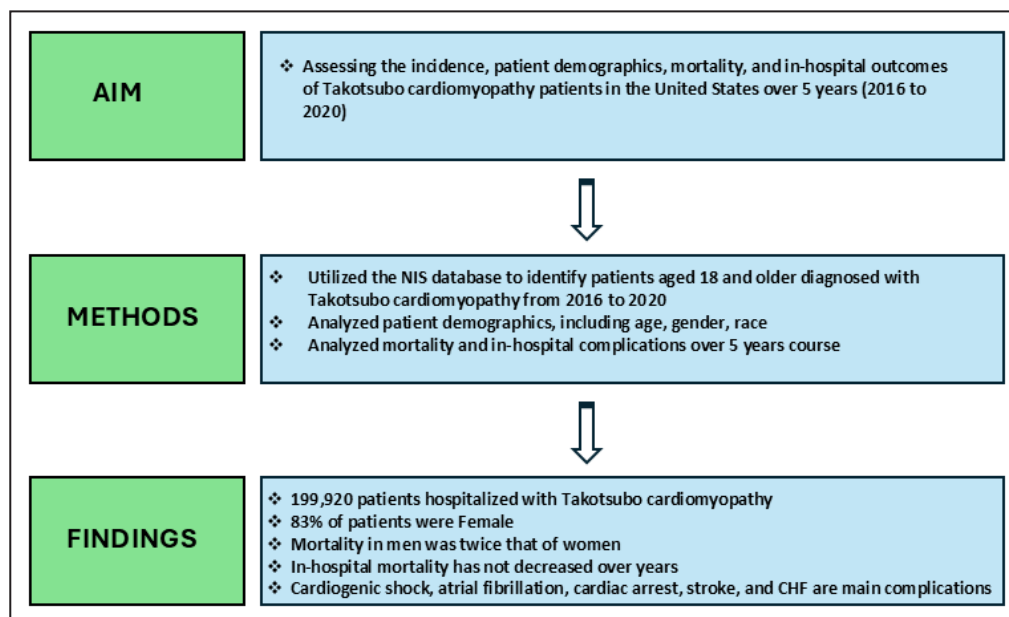


Figure 6. Visual illustration of aim, methods, and major findings.

CHF indicates congestive heart failure; and NIS, Nationwide Inpatient Sample.

in 2020. This >1.5% increase in mortality from 2019 to 2020 can be attributed to multiple factors. Physical stress associated with TC is becoming more focus of attention and there are currently more studies investigating its incidence and prognosis.²³ The mortality rate resulting from TC related to physical stressors is higher.^{23,24} One of the most important recently discussed physical stressors is COVID-19.²⁵ COVID-19 emerged in late 2019 and multiple studies reported significant increases in TC due to this pandemic.^{26,27} In the review by Chang et al., the inpatient mortality of patients with COVID and TC is higher than TC alone.²⁸ So, it seems that COVID might play a role in increasing the incidence and mortality rate in recent years.²⁷

Patients hospitalized with TC had substantial rates of other cardiovascular complications. Complications increased over the years, except for atrial fibrillation and myocardial rupture, which had a steadier course. Commensurate with other studies, CHF was the most common complication of TC¹⁶ followed by atrial fibrillation. In a study by Templin et al., in hospital complications and death were similar to patients with acute coronary syndrome.²⁹ In another study by Vallabhajosyula et al., patients with TC had a higher risk of heart failure and lower risk of mortality compared with myocardial infarction during the 8-year period from 2007 to 2014.³⁰ Early diagnosis of stress induced cardiomyopathy can be done using classic echocardiographic features³¹ that may help to recognize and treat this condition earlier before serious complications such as stroke can occur. These data confirm that complications following TC are still high and even have incremental patterns over the years, mandating more research and intervention in this field.

One of the most important findings in our study is the double mortality rate of male patients presenting with TC. The cause of this discrepancy in mortality is not fully understood. This fact has been observed in previous studies. Arcari et al.³ found that mortality in men was 10% versus only 3.8% in women. After multivariate adjustment, men had more than twice the risk for mortality. A recent meta-analysis confirmed a double mortality rate in men.³² In a large TC registry, men had more physical stress as a trigger event and cardiac arrest as presenting symptoms that could be one explanation for higher mortality^{33,34}. Hormonal differences with higher levels of catecholamines in men can also play an important factor.³⁵ Future studies are required to study the reason behind higher mortality in men.

Strengths of this Study

There are several strengths to our study. The present study is one of the largest studies on TC assessing race or ethnicity, sex, and age differences along with a wide

variety of complications and in-hospital mortality in a 5-year study course. Using nationally representative data allows the generalizability of the findings and eliminates the biases associated with small sample sizes and single-center studies.

Limitations

The data were obtained from NIS using ICD-10 codes. As a result, they are prone to coding errors. We did not have data on different types of TC and the incidence of each subtype, hence associated outcomes with each type cannot be evaluated. NIS is an inpatient data register and did not have access to outpatient data, therefore long-term mortality cannot be evaluated. Another limitation is the lack of data on patients' comorbid conditions that might contribute to their mortality. Furthermore, an NIS entry is equivalent to 1 hospitalization. If a patient was hospitalized more than once, then 1 patient could have multiple entries within the study period. Furthermore, if a patient had an event in one hospital and then was transferred to another hospital, then it is possible that the same case may have been recorded more than once. Additional studies are warranted to mitigate these limitations.

CONCLUSIONS

The data clearly indicate that individuals with Takotsubo syndrome are at significantly higher risk for a range of serious cardiovascular complications. Men have more than twice the mortality in comparison in women. Furthermore, there has been no improvement in any of the complications of interest throughout the years of the study. Further research on management and improvement in the care of these patients is warranted.

ARTICLE INFORMATION

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Disclosures

None.

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